



Using inequality notation for errors in calculations

1 An error interval for a number x shows the range of possible values of x . It is written as $a \leq x < b$. Explain what this means. Tick **1** correct answer

- ☐ The range of values include a and b
- ☒ The range of values include a and not b
- ☐ The range of values includes all the numbers strictly in between a and b
- ☐ The range of values do not include a but include b

2 When a number, x , is rounded to the nearest integer the result is 63. Select the numbers which complete the error interval for x : $\square \leq x < \square$ Tick **1** correct answer

- ☐ 62.5 and 65.5
- ☐ 63.5 and 64.5
- ☐ 62.5 and 64.5
- ☒ 62.5 and 63.5

3 When a number, x , is rounded to the nearest integer the result is 5. Select the numbers which complete the error interval for x : $\square \leq x < \square$ Tick **1** correct answer

- ☐ 4.5 and 5.6
- ☒ 4.5 and 5.5
- ☐ 3.5 and 6.5
- ☐ 4.5 and 5.49

4 When a number, x , is rounded to the nearest integer the result is 10. Select the numbers which complete the error interval for x : $\square \leq x < \square$ Tick **1** correct answer

- ☐ 9.5 and 10.4999
- ☐ 5 and 14.5
- ☐ 5 and 14.49999
- ☒ 9.5 and 10.5



- 5 Jun rounds a number to the nearest integer and the result is 5. Izzy rounds a number to 1 d.p. and the result is 5.0. Jun says, "We must both have rounded the same number." Is Jun correct? Tick **1** correct answer
- ☐ Jun is correct as both are the same when rounded to 1 d.p
- ☐ Jun is correct as both values will be the same when inputting into a calculator.
- ☒ Jun is incorrect as they have different error intervals.
- 6 x and y are both integers. $x = 650$ (rounded to the nearest 10) and $y = 800$ (rounded to the nearest 100). The greatest possible integer value of $x + y$ is 1503. Fill in the blank

